

mineralization pyrite galena
granite apatite hematite
garnet fluid inclusions chalcopyrite
sphalerite quartz calcite
biotite chlorite magnetite
zircon olivine plagioclase sulfides
pyrrhotite

delta34S gold deposits ore bodies
delta18O
tourmaline dolomite Se Au mesozoic Ta Sb Hf TiO2 metamorphism Pd
W LREE F scheelite Zr H2O Ca SiO2 Sn S
feldspar Cr Sr K-feldspar MgO Ni
magmatism Na Fe Na Co Zn Nb CO2 fractures silicification molybdenite ilmenite
hornblende albite PGE Ag Pb HREE chromite amphibole Ba
PGE Ag Pb HREE chromite
fluorite zircon titanite sericite cassiterite Al
REE barite
faults Bi Y
hydrothermal alteration carbonates
hydrothermal fluids
volcanic rocks
muscovite
pentlandite
granodiorite
Rb
partial melting Al2O3 Mn
phlogopite
orthopyroxene
Mo
Eu
Th
uranine
Si
alteration
granitoids
pyroxene
V
epidote
Ni
arsenopyrite
clinopyroxene
quartz veins
rutile
monazite

ore-forming fluids
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